Technical Session on Bamboo
DUDBC Conference Hall, 29th November 2016

<u>Participants</u>: GOAL, Samaritan's Purse, Shock Safe Nepal, Helvetas, Concern, Himalayan Bamboo, CRS, UNOPS, Japanese Red Cross, Abari, and Habitat for Humanity.

<u>Introduction</u>

The HRRP technical sessions are designed for people working on a particular topic to share their experiences and challenges, and for others to have a chance to hear what is happening on that topic. The session on bamboo brought key actors together to discuss technical challenges and achievements that they have been dealing with as part of the reconstruction.

<u>Presentation on Bamboo, Kishor Neupane, Habitat for Humanity Nepal</u>

Habitat for Humanity has been working on bamboo cultivation, treatment, and promotion as a housing material since 2006. The main purpose of promoting bamboo is that it's environmentally friendly and low income households can afford it. Habitat for Humanity have established 42 nurseries in Jhapa, Morang, Sunsari, Saptari and Dang districts and have conducted training on nursery setup and management and bamboo cultivation. If bamboo is now promoted for the reconstruction process and used at scale for housing then there will be a shortage of bamboo also so it is important to consider the fully supply chain.

Habitat for Humanity work through local partners in the districts particularly Community Forest User Groups (CFUGs), women's groups, cooperatives, etc. When Habitat for Humanity first introduced bamboo in 2006 it was very difficult to get people to adopt bamboo for housing, but now following training and experience, and seeing examples that have been constructed, people are slowly beginning to adopt the material in their house construction. People have also found that bamboo cultivation is a good source of income generation. Habitat for Humanity have also trained mason experts on bamboo, which is another livelihood aspect.

The treatment process which Habitat for Humanity use involves submerging the bamboo in a mixture of boric acid and borax for two weeks. The treatment is conducted in Morang and Jhapa districts. Designs have been developed and implemented as demonstrations. The first design, ten years ago, was well received as it is light and affordable but there were problems with the design due to moisture in the bamboo column because of the connection to the concrete. The connection was changed to a metal strap set in the concrete which then connects to the bamboo column. A new design was also developed with an overhanging roof to protect the bamboo from sun and rain. Bamboo is also being promoted as a roofing material as a cheaper alternative to timber.

Habitat for Humanity are now planning to work more on the market chain aspects of bamboo to link producers and end users. Habitat for Humanity are planning research on this and they will be investing more in this process. Habitat for Humanity are keen to collaborate with any other partners that are interested in working on the market chain development aspects of bamboo. The contact details for Habitat for Humanity are included in the 'contact details' section below.

Q: What is the ratio of treatment materials, and what quantities need to be used?

- The ratio is one part boric acid to 1.5 parts borax, and 100 parts water. The tank used is 30 x 6 x 3 ft. and can fit 200 hundred lengths of bamboo and these are treated for a minimum of 15 days.

Q: How long does bamboo last for after it has been treated using this process?

- From 10-50 years, depending on the bamboo species and the treatment process. There are some bamboo buildings that are hundreds of years old in other countries with similar climates.

Q: What climatic conditions are required for growing bamboo?

- Bamboo is mainly grown in the Terai region, but there are species of bamboo that grow up to altitudes of around 1500 m and Nepal has a particularly rich mix of bamboo species grown throughout the country due to the different climates found in the different belts of the country.

Sustainable Building / Creating Livelihoods, Nripal Adihikary, Abari

Abari was established in 2006 with the aim of taking traditional materials and using them in a modern context. The research and design firm is based in Chitwan, and has completed projects in India, Bhutan, Ethiopia, Kenya, Uganda and USA, in addition to Nepal. Abari work on the full bamboo supply chain looking at five key areas; farming and documentation, pre-processing, research and development, production, and training.

In the past five years, Abari have developed a bamboo plantation on a plot of roughly 2 hectares in Chitwan. The plantation includes 24 species of bamboo collected from all over Nepal. A mobile app has also been developed to track the location of bamboo cultivation by GPS. The app includes contact details, information on species grown, number of clumps, etc. It has been piloted in Morang, Jhapa, and Kavre

Tubes are one of the strongest structures that exist and bamboo is full of tubes. Bamboo needs sugar to grow and insects can attack the sugar held inside bamboo, or bamboo can decay. This is why treatment is required when using bamboo as a structural element. Bamboo is treated to remove the sugar, and a traditional way of doing this is to place the bamboo in water, preferably running water, for a minimum of three weeks to dissolve the sugar. Abari have also developed a boucherie treatment system which uses pressure to push the sugar out of the bamboo and replace it with boric acid or another preservative. A mobile boucherie treatment service can be very effective if it is well managed, but bamboo must be treated immediately after cutting. An example of a mobile treatment system was shown where the compressor and other treatment materials are attached to a bike and the treatment costs 5 NPRs per length of bamboo. Abari also have an industrial treatment plant in Chitwan using a pressure chamber, which is a standard treatment system for timber also.

Abari have developed and are piloting a portable grading machine to test the strength of bamboo without destroying it. The machine uses a car jack to apply pressure and look at displacement in the bamboo. Testing was also carried out in the Institute of Engineering (IoE) on the steel and bamboo joinery system which Abari has developed. The system allows for standardised joints and the testing in IoE found that the steel breaks at 3.5 tonnes, but the bamboo was not affected. A bigger steel plate had to be used to see where the bamboo breaks. A video of the testing was shown illustrating this. A simple compression testing machine is used in the field to show the crushing strength for bamboo or soil blocks. This way the knowledge becomes more accessible to households and masons and the perceptions of materials can start to change. It is equally important to convince communities as it is to convince policy makers that materials such as bamboo and earth can be used for housing construction. Abari are working with Kathmandu University to create a material testing laboratory. The laboratory is being constructed with bamboo and earth, and is almost complete. The laboratory is also expected to engage students in research.

Since the earthquake, Abari have had a bamboo and earth design for schools approved by the GoN as a type design. This takes a standard design and makes it more sustainable in terms of the materials used. The design also includes considerations for sound and thermal insulation and air flow. Bamboo is used for the roof because it is light weight, insulating, looks good, costs less than timber, and is locally available. The investment also stays with the local community. Abari have found that when you build local materials, approximately 89% of the investment stays locally. Some materials have to be brought in (steel, cement, etc.) but these are kept to a minimum.

Q: Is the Abari school type design freely available for others to use?

- Yes, it is open source and is available on the Abari website and Department of Education website.

Q: Does using mud mortar with bamboo structures effect the strength of bamboo? How long will the buildings last?

- No, mud mortar has no effect on the strength of bamboo. The buildings will last at least 50 years, but should last forever where none of the bamboo is exposed. Treatment is very important but it's also really important to design out exposure of the bamboo. But if there are any parts that become damaged or worn, it is possible to change individual pieces.

Q: How can bamboo be treated locally?

- Bamboo should never be cut in a wet climate as it increases the risks of insects attacking it. Traditionally local treatment involves placing the bamboo in water for an extended period of time to dissolve the sugar held within it. It is however recommended that more improved treatment be used where bamboo will be used structurally. There are a number of different treatment options available, and it is important to assess which is best for the location and scale of the project.

Q: If using the traditional method of treating bamboo in water, how long is required for treatment?

- A minimum of three weeks to one month and it is better if running water is used.

Q: What sort of joints does Abari use for low cost housing? Do you think households will adopt these types of joints?

- Steel bands are used with two holes and these connect to the bamboo. It is expected that households will adopt these joints as they are not heavily engineered and can be easily replicated.

Q: What is total cost of the steel used in the low cost house?

- The total cost of the house was 8 lakh and the house has three rooms. The cost for the steel joints was not very high and the joints are very simple and can be made by any welder.

Q: What is the cost comparison between a bamboo truss and a steel truss?

- Normally a bamboo truss will be about 15% less expensive than a steel truss.

Q: In the Terai where people use bamboo for walls, traditionally mud plaster was used but people are now moving to cement plaster. What is the life span when cement plaster is used?

- Mud plaster is preferred to cement as mud regulates and breathes but cement is hydroscopic and sucks in water from the bamboo so they will separate at some point. Using mud is also a way of preserving bamboo as it reduces the moisture content so insects don't thrive. Lime plaster is also a good option.

Q: Can Abari supply bamboo for housing construction?

- Abari can supply, but would prefer to have a more sustainable supply chain developed. If you were putting load vertically along the fibre of bamboo it doesn't matter which species is used, but it is best to avoid long pieces in the design because they can buckle. There is lot of bamboo in Nepal but cultivation is generally disorganised. Abari want to use locally available and develop designs based on the locally available species.

Q: Does Abari have any plan to scale up and support other districts, especially those affected by the earthquake?

- Abari works in Dhulikhel, Kavre and are focused there because of limited capacity to spread further into earthquake affected areas. Work is also continuing outside of the earthquake affected districts in the Terai.

Q: If there are community members who would like training on bamboo can you support them?

- Yes, there is a training centre in Dhulikhel, Kavre where people can come and participate in training.

CRS Approach on Bamboo, Suraj Parajuli and Nischal Pradhan, CRS

CRS began working on bamboo approximately 13 months ago, starting with a collaboration with Humanitarian Benchmark Consulting (HBC). CRS's approach to bamboo is focused on six key areas:

- Effective treatment machine,
- Construction techniques,
- Research on pests,

- Demonstration house design using bamboo,
- Pilot workshop in the community, and
- Development of a bamboo training manual.

When CRS started conducting community workshops on bamboo they found that people have a lot of knowledge on bamboo that has been passed down from their ancestors but additional information and support was also required, especially on treatment and use of bamboo in house construction. An information leaflet, in the same format as the Shelter Cluster 10 key messages posters, was developed for the full life cycle of bamboo through to design and construction. CRS have found that people are really interested as nearly everyone has bamboo, and they are keen to get more information on how to treat and use it. HBC provided a one day ToT to the CRS team on the community orientations and also produced facilitation guidelines for cultivation and harvesting orientation sessions.

Treatment: CRS initially started looking at using a boucherie treatment machine that included a metal compressor. The idea was to have a treatment system that could be transported through villages providing bamboo treatment as a service. Because the metal compressor is very heavy and many of the metal parts could not be sourced locally CRS reviewed the approach and came up with a gravity boucherie treatment system. The gravity system makes use of the topography of the area, and removes the need for the metal compressor. All metal parts were replaced with plastic plumbing fittings, and all the materials used in the system can be easily sourced locally. All the components can fit in the bucket (which is used for the treatment chemicals) for transport. The treatment chemicals used are boric acid and borax. The information pamphlet for the system is available upon request from Adeel Javaid (Adeel.Javaid@crs.org).

Pest Research: CRS found that every clump they visited had a very high level of pests present, with up to 90% of plants in a clump were infested. A survey was conducted to gather more information on the pest – the 'shoot borer moth'. In the villages where the survey was carried out many people in the community had some knowledge on the pests but it was found that the pests are not addressed, people simply use only the bamboo that is not infested. The pests were found to be most prevalent around 1,000m and below 700m there was a significant reduction in the presence of pests. The life cycle of the moth is such that it makes a hole in the bamboo to enter, then eats the cavities of the bamboo, eventually turns into a moth and leaves the bamboo through the hole it entered through. The moth reduces the size of the internode and the strength of the bamboo. There is very little research on this but CRS did find one paper written by Stapleton, but found that there were some differences between the findings from the CRS pest survey and the information presented in the paper. A pest control guide has been developed and is available here.

Treatment Test Kit: Developed in order to be able to test bamboo to know if it has been treated or not. Initial idea was to use litmus paper but this did not work. Found that one spoon of turmeric powder mixed with 9 spoons of ethanol can be used to indicate whether bamboo has been treated or not. When bamboo has been treated the colour of the mixture will change to red indicating that boron present. If the bamboo has not been treated the colour will remain yellow.

Demonstration House Design: CRS have developed a demonstration house design which has a stone masonry and mud mortar, with RCC banding, ground floor and a bamboo attic and roof. The foundation is cement mortar masonry as mud mortar masonry should not be used for foundations as it absorbs water and the strength reduces quickly. The walls of the attic, and the gables are made using bamboo strips infilled with a thin layer of mud. The roofing material used is CGI sheets. The demonstration house design is still going through the approval process with DUDBC and this has been quite challenging as there have not been any housing designs using bamboo approved previously (either before the earthquake or after). The basis of structural design for RCC, steel, and bamboo structures were compared, indicating the relatively weak engineering knowledge base for bamboo as compared to RCC and steel. A comparison of the material properties of concrete, structural steel, and structural bamboo (B.balcooa) was also presented and the modulus of elasticity indicates that the flexibility of bamboo is high and must be taken into account when designing. CRS prepared a full structural analysis of the design as part of the process to secure approval from DUDBC, and one of the challenges is that there is no software that specifically deals with bamboo so the joints need to be calculated by hand using structural engineering principles. The joints must include appropriate elements to take care of the critical stresses indicated by the structural analysis.

Q: Is there any information available on the thermal expansion properties of bamboo?

- The coefficient of thermal expansion for bamboo ranges:
  - From 2.56 to 4.12 *10^(-6) per degree Celsius with standard deviation of 0.27 to 1.17 parallel to grain (depending upon test method), and
  - From 42.66 to 54.54 *10^ (-6) per degree Celsius with standard deviation of around 3.8 across the grain (again depending upon test method).
- More information can be found here.

Q: In which districts is it feasible to use bamboo for reconstruction?

- It is feasible in all districts but supply / market chain has to be really considered and support provided where needed.

Q: How does CRS reach out to communities with information on bamboo and is literacy rate an issue?

- CRS use a combination of pictures, practical sessions, and community orientations so there is no reliance on written materials and literacy rate is not an issue.

Q: Is the gravity boucherie treatment system available to purchase?

- It is not available to purchase as a set but an information pamphlet is available upon request from Adeel Javaid (Adeel.Javaid@crs.org), and this provides the details on how to put together the treatment system.

Q: What can the maximum length of a bamboo beam be?

- This should be decided based on the amount of deflection the beams will be required to sustain and what level of deflection is considered acceptable in the design. The beam size must meet the design requirements considering the stress the beam will be subject to. Buckling column theory should be consulted to determine the length allowed based on the cross section of the column.

Rebuilding a Sustainable Nepal, Akriti, Himalayan Bamboo Pvt. Ltd.

Himalayan Bamboo is a private company which has been working on bamboo construction and furniture production since 1995. The reasons why bamboo is considered a viable housing option were presented in detail; strength, higher earthquake resistance, energy efficiency, durable with proper treatment, grows widely, densely, and quickly, potential carbon credit, and renewed local livelihoods (supply chain). The challenges that Himalayan Bamboo have faced to date in using bamboo for housing include social acceptance, absence of national standards for bamboo housing, if a bamboo house is constructed in one location it cannot be shifted to another (when the Bahareque technique is used), and the unsystematic approach to bamboo in terms of cultivation, treatment, and use. Examples of Himalayan Bamboos work were shown, including schools and health posts.

Himalayan Bamboo uses copper chrome boron treatment with an anti-termite chemical treatment also. A solution of 6-8% is used to treat bamboo for indoor use, and a solution of 8-10% is used to treat bamboo for outdoor use.

The key parts of the bamboo supply chain were presented; farmers from village (cultivating bamboo), harvesters, primary processing, supplier, transporters, manufacturer, trader, retailers, and customers.

Himalayan Bamboo have constructed a model house in Bhaktapur prior to the earthquake using ‘Bahareque’ technique, a technique which INBAR supported Himalayan Bamboo to develop. In this technique the foundation is constructed from stone masonry, then a 2” timber frame makes up the walls, and strips of bamboo are laid horizontally across the timber frame. The walls are then covered in GI wire mesh and cement plaster is applied on both faces. The bamboo and timber walls were assembled at Himalayan Bamboo’s factory in Hetauda and transported to the site for installation. The foundation is 1 ft. below ground with a plinth 1 ft. above ground. GI wires are set into the foundation and the wall panels connect to the foundation using these. The house has two bedrooms, a toilet, and a kitchen and is 360 sq. ft. in area. They include all sanitary, drainage, and electrical requirements and can use the same approach to construct commercial buildings also.

A resource comparison was carried out for the 360 ft. model house design and it was found that if it was constructed in RCC it would cost 8-12 lakh rupees, but when it was constructed in bamboo it cost 8 lakh rupees including furniture and could be constructed for 5 lakh rupees.

Q: How much did the model house in Bhaktapur cost to construct?

- It cost 8 lakh NPRs, including furniture. It was completed just before earthquake so the cost may be higher now.

Q: Has DUDBC approved the model house design?

- No, Himalayan Bamboo have not secured approval from DUDBC for the model house design.

Q: Are there any security issues with this type of house given the walls are quite thin?

- No, there are no security problems. The bamboo strips used in the wall are very strong. Anyone interested to see how the structure looks are welcome to visit the Himalayan Bamboo office in Thapathali which has been constructed using the Bahareque technique.

Q: Where is the model house in Bhaktapur located?

- It is located on the way to Nagarkot, on the main highway.

Q: What measures have been taken in the design to prevent fire?

- Cement plaster is used on both sides of the walls to protect the bamboo and timber from moisture and fire. The life span of the house is estimated to be 30-40 years as the bamboo is treated and cement plaster is used on both sides. INBAR has provided Himalayan Bamboo with research papers which refer to similar style houses in South America that are 100+ years old. As Himalayan Bamboo has only been working since 1995 the estimate is kept to 30-40 years.

Q: How thick are the walls?

- The walls are made up of a 2 inch timber frame, plus the bamboo strips, plus 1.5 inch plaster on both sides so the walls are roughly 6-7 inches thick.

Next Steps

- The HRRP will continue to collaborate with the presenters, and all interested partners, to compile existing testing and research and identify additional testing and research needs. It is hoped that by building up the base of engineering data on bamboo the process to approve housing designs using bamboo will become easier.
- The HRRP will advocate on behalf of partners and households for stronger promotion of bamboo as an alternative to timber for the housing reconstruction.

Reference Documents

- Pest Control Guide developed by CRS, available here
- Pest Survey Report from Gorkha district prepared by CRS, available here
- Bamboo Cultivation Workshop Facilitation Guide developed by CRS, available here
- Bamboo Harvesting Workshop Facilitation Guide developed by CRS, available here
- Bamboo information posters developed by CRS, available here
- HBC Report 1, Bamboo in Nepal, available here
- HBC Report 2, available here
- HBC Report 3, available here
- HBC Report 4, available here

- HBC Report 5, available here
- Bamboos of Nepal, Chris Stapleton, available here
- Mechanical Properties of Bamboo, Jules J.A. Janssen, available here

Contact Details

Contact details for the teams presenting at the technical session are provided below in order to facilitate follow up on any of the topics covered in the session.

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